

Aim: Multithreading.**1. Write a java program to create thread using Thread class.****CODE:**

```
class NewThread extends Thread{
    NewThread(){
        super("Demo");
        System.out.println("My child thread in constructor: " + this);
        start();
    }
    public void run(){
        try{
            for(int i=5;i>0;i--){
                System.out.println("Child thread value:" + i);
                Thread.sleep(1000);
            }
        }
        catch(InterruptedException e){
            System.out.println("Child thread is interrupted!");
        }
        System.out.println("Child thread is finished!");
    }
}

public class CreateThread {
    public static void main(String args[]){
        new NewThread();
        try{
            for(int i=10;i>0;i--){
                System.out.println("Main thread value:" + i);
                Thread.sleep(1500);
            }
        }
        catch(InterruptedException e){
            System.out.println("Main thread is interrupted!");
        }
        System.out.println("Main Thread finished!");
    }
}
```

OUTPUT:

```
My child thread in constructor: Thread[#35,Demo,5,main]
Main thread value:10
Child thread value:5
Child thread value:4
Main thread value:9
Child thread value:3
Main thread value:8
Child thread value:2
```

Child thread value:1
Main thread value:7
Child thread is finished!
Main thread value:6
Main thread value:5
Main thread value:4
Main thread value:3
Main thread value:2
Main thread value:1
Main Thread finished!

2. Write a java program to create a thread using Runnable Interface.**CODE:**

```
class NewThread implements Runnable{
    Thread t;
    NewThread(){
        t = new Thread(this, "Demo Thread");
        System.out.println("My child thread in constructor:"+t);
        t.start();
    }
    public void run(){
        try{
            for(int i=5;i>0;i--){
                System.out.println("Child thread value:"+i);
                Thread.sleep(1000);
            }
        }
        catch(InterruptedException e){
            System.out.println("Child thread is interrupted!");
        }
        System.out.println("Child thread is finished!");
    }
}

public class RunnableThread {
    public static void main(String args[]){
        new NewThread();
        try{
            for(int i=10;i>0;i--){
                System.out.println("Main thread value:"+i);
                Thread.sleep(1500);
            }
        }
        catch(InterruptedException e){
            System.out.println("Main thread is interrupted!");
        }
        System.out.println("Main thread is finished!");
    }
}
```

OUTPUT:

My child thread in constructor: Thread[#35,Demo,5,main]

Child thread value:5

Main thread value:10

Child thread value:4

Main thread value:9

Child thread value:3

Main thread value:8

Child thread value:2

Child thread value:1

Main thread value:7

Child thread is finished!

Main thread value:6

Main thread value:5

Main thread value:4

Main thread value:3

Main thread value:2

Main thread value:1

Main thread is finished!

3. Write a java program to demonstrate synchronization of threads.**CODE:**

```
class Counter {
    int count = 0;
    synchronized void increment() {
        count++;
    }
}

class MyThread extends Thread {
    Counter c;
    MyThread(Counter c) {
        this.c = c;
    }
    public void run() {
        for (int i = 0; i < 1000; i++) {
            c.increment();
        }
    }
}

public class SynchronizedThread {
    public static void main(String[] args) {
        Counter counter = new Counter();
        MyThread t1 = new MyThread(counter);
        MyThread t2 = new MyThread(counter);
        t1.start();
        t2.start();
        try {
            t1.join();
            t2.join();
        }
    }
}
```

```
    }  
    catch (Exception e) {  
        System.out.println(e);  
    }  
    System.out.println("Final count:" + counter.count);  
}  
}
```

OUTPUT:

Final count:2000

WITHOUT SYNCHRONIZED:

Final count:1990